

How can we increase revenue from Catch the Pink Flamingo?

Analyzing user behavior across sessions, purchases, and social networks to unlock monetization opportunities for Eglence Inc.

CAPSTONE PROJECT REVIEW

EGLENCE INC.

Project Goal, Data Sources, and Methods

Objective

Analyze user behavior to improve monetization and engagement across the Catch the Pink Flamingo platform.

Data Sources

- User sessions and game clicks
- Ad clicks and purchase events
- Team assignments and chat interactions



Splunk

Descriptive statistics — platform usage, ad performance, purchase trends



KNIME

Decision tree classification — buyer segmentation



PySpark

K-Means clustering — behavioral profiling



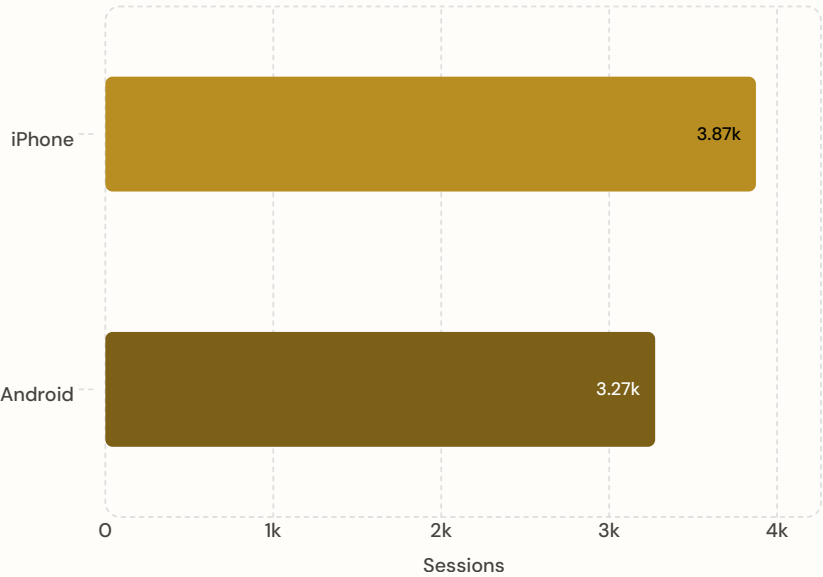
Neo4j

Graph analytics — social network structure

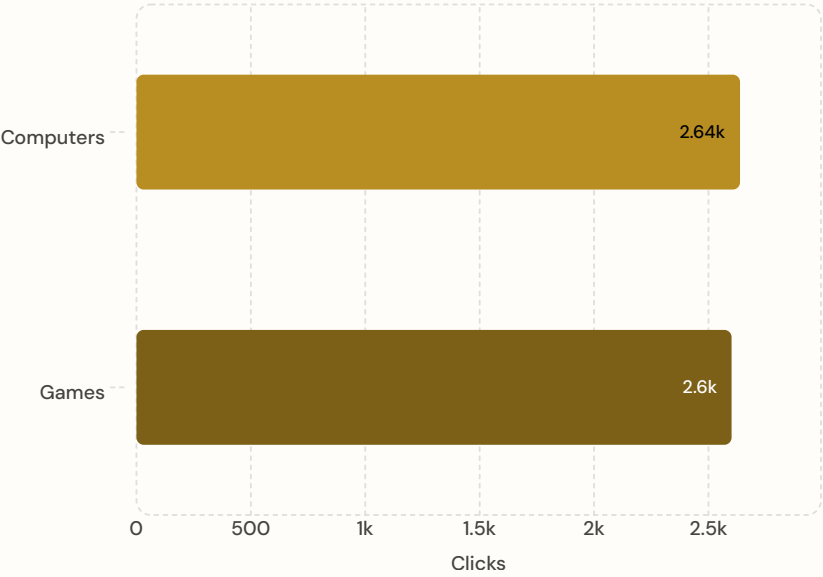
Week 1: Platform Usage and Ad Performance

SPLUNK

Platform



Ad Category



Key Findings

- **iPhone** leads with 3,874 sessions; **Android** follows with 3,274 — mobile-first audience confirmed
- **Computers** (2,638 clicks) and **Games** (2,601 clicks) dominate ad engagement
- **Item 2** (714 purchases) and **Item 5** (610 purchases) are the top-selling in-game items
- Average team size of **78 players** signals strong collaborative gameplay culture

📌 Mobile platforms account for all top sessions — any monetization strategy must be mobile-optimized.

Week 2: Buyer Segmentation

KNIME CLASSIFICATION

A decision tree model classified **1,411 paying users** into two distinct buyer types based on average transaction spend.

PennyPinchers

Average spend $\leq$ **\$5.00** per transaction

Price-sensitive buyers who respond to low-cost entry points and value offers.

HighRollers

Average spend $>$ **\$5.00** per transaction

Premium buyers with higher willingness to pay — candidates for exclusive content and loyalty programs.

Key predictors: **team level**, **platform type**, and **engagement counts**. User IDs and the target variable were excluded to prevent data leakage and ensure model integrity.

Week 3: Behavioral Clustering

PYSPARK K-MEANS

Casuals 80% · n=1,912

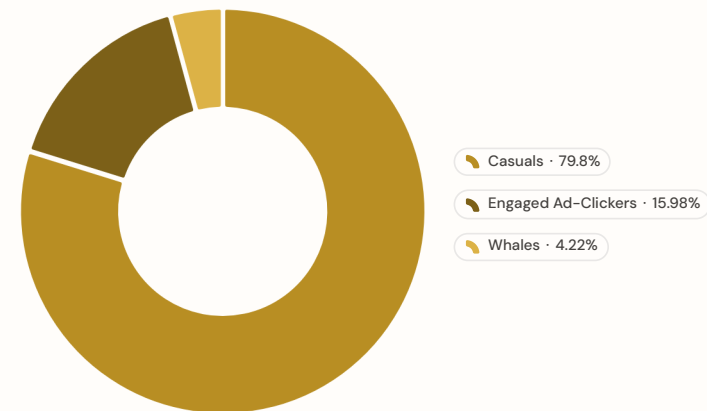
150 game clicks · 28 ad clicks · **\$0.83 avg. revenue**
Low engagement, minimal spend. Largest segment by volume.

Engaged Ad-Clickers 16% · n=383

1,020 game clicks · 28 ad clicks · **\$22.75 avg. revenue**
High play volume, moderate spend. Strong ad interaction.

Whales 4% · n=101

805 game clicks · 40 ad clicks · **\$110.59 avg. revenue**
Premium engagement and spending. Highest value segment.



Clustering basis: total game clicks, ad clicks, and revenue across **2,393 users**.

Week 4: Social Network Structure

NEO4J GRAPH ANALYTICS

45K

Nodes

Unique users mapped in the network

118K

Relationships

Chat interactions between users

1.0

Clustering Coefficient

Top users show perfect cohesion — every member interacts with every other

What the Graph Reveals

- The network spans **45,463 nodes** and **118,502 relationships**
- Longest conversation thread: **10 messages** across 9 replies involving 5 users
- Hyper-active users cluster within hyper-active teams — social behavior is self-reinforcing
- Perfect clustering coefficients (1.0) identify tightly-knit communication pods

📌 Social cohesion *correlates* with sustained engagement and spending — causation requires further study.

Monetization Recommendations

\$0.99 Starter Packs → Casuals (n=1,912)

Converting even a small fraction of 1,912 dormant users generates significant volume. A low price point reduces friction for a first purchase and opens the monetization funnel to the largest segment.

Premium Ad Placements → Engaged Ad-Clickers (n=383)

This segment plays heavily and clicks ads but spends moderately on direct purchases. Scaling ad revenue through premium placements captures value without cannibalizing in-app sales.

VIP Loyalty Tier → Whales (n=101)

High spenders averaging \$110.59 require exclusive rewards and recognition to sustain their spend and prevent churn to competitors. Retention here protects the highest-value revenue stream.

Conclusion: From Data to Strategy



Splunk

KNIME

PySpark

Neo4j

This roadmap enables Eglence Inc. to monetize distinct segments differently based on concrete usage data — moving from one-size-fits-all pricing to **targeted engagement strategies** grounded in evidence from four complementary analytical methods.